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Accuracy and reliability of automated three-dimensional facial landmarking in medical and biological studies. A systematic review

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Title: Accuracy and reliability of automated three-dimensional facial landmarking in medical and biological studies~~.~~ : A systematic review

Abstract

Background

3D facial landmarking is becoming a fundamental part of clinical and biological applications. Manual landmarking is time-consuming and prone to cumulative errors~~. A~~, so attempts ~~were~~ have been made to automate 3D facial landmarking~~, h~~. However, data in the literature ~~is~~ are sparse.

Objectives

~~To~~ The objectives of this study are to investigate ~~the~~ current evidence ~~of~~ for the accuracy and reliability of various 3D facial automated landmarking methods used in medical ~~or~~ and biological studies and evaluate their performance against the manual annotation method.

Search methods

Electronic and manual searches of the literature were performed in April 2021.

Selection criteria

Only studies that were published in English and evaluated the accuracy of automated landmarking algorithms in 3D facial images for medical or biological settings were included.

Data collection and analysis

Two authors independently screened the articles for eligibility. The QUADAS-2 tool was used for the quality analysis of the included studies. Due to the heterogeneity of the selected studies, a meta-analysis was not possible, ~~and~~ so a narrative synthesis of the findings was performed.

Results

From 1002 identified records, ~~and~~ after applying the inclusion and exclusion criteria, 14 articles were ~~finally~~ ultimately selected, read, and critically analysed. Different algorithms were used for the automated 3D landmarking of various numbers of facial landmarks ranging from 10 to 29 landmarks. The average difference between the manual and automated methods ranged from 0.67 to 4.73 mm, and the best performance was achieved ~~by u~~using deep learning models. Poor study design and inadequate reporting were found in the implementation of the reference standards and population selection for the intended studies, ~~and this~~ which could have led to overfitting of the tested algorithm.

Conclusion

Compared to manual landmarking, automated Landmark localisation of individual facial landmarks reported in the literature is not accurate enough to allow their use for clinical purposes. This result indicates that automatic facial landmarking is still developing, and further studies are required to develop a system that could match or exceed the performance of the current gold standard.

Registration

PROSPERO: CRD42021241531.

Conflict of interest

None to declare.

Background:

Three-dimensional (3D) facial images are routinely used to measure soft tissue characteristics in medical settings and biological research to accurately and comprehensively quantify facial morphology (1, 2). In medicine, the use of 3D imaging provides planning and outcome measures of orthognathic surgery, by evaluating the soft tissue changes following ~~the~~ surgery (3).

In genetic and developmental studies of facial morphology, using large datasets, 3D facial images integrated with genomics can unravel the genetics of facial morphological shape variation. This technology has enabled deep phenotyping, which allows ~~for studying~~one to study the genomic architecture of facial shape (4, 5). Provided the level of accuracy is acceptable, the objectivity, consistency, and efficiency of digital computation ~~is~~ represent a significant advancement, mainly when dealing with large data sets.

Well-defined anatomical landmarks over the face allow the shapes to be characterised ~~in a manner that corresponds~~ across subjects and, therefore, provide the basis of subsequent statistical analysis. Accurate placement of these landmarks on ~~the~~ 3D facial scans is essential, and the results of any further analysis depend on the accuracy of the identification and localisation of ~~those~~ such landmarks. Manual landmark identification is time-consuming and prone to cumulative errors ~~and time-consuming~~, especially when carried out on large datasets; therefore, robust and accurate automated 3D facial landmarking tools are highly desirable.

Various automated techniques for detecting anatomical facial landmarks have been reported (6-8). To our knowledge, there have been no systematic reviews of the automated 3D facial landmark technique performance compared to the manual method in medical and biological settings.

The purpose of this systematic review is to examine ~~the~~ current evidence regarding the accuracy and reliability of various 3D facial automated landmarking methods used in medical ~~or~~ and biological studies and evaluate their performance against the gold standard expert manual annotation method. This review ~~will p~~provides a comprehensive and contemporary overview, identify identifies gaps in the literature, and establishes a baseline for future research.

~~The~~ This systematic review aims to answer the following questions:

1) How accurate is automated 3D facial landmarking compared to manual landmarking, and which facial landmarks are more or less reliable?.

2) What is the reliability of the soft tissue facial analysis obtained through automated landmarking compared to ~~those~~ analysis based on manual landmarking?.

3) What are the possible gaps in automated 3D facial landmarking in clinical and biological research?.

In essence, it is highly desirable that, in this field of digital imaging, we succeed in producing an accurate automated alternative to manual facial landmarking that is also feasible, practical, and user-friendly, and ~~that~~ can, through a user interface, ~~can~~ be rolled out across the world for clinical governance and research purposes.

Materials and methods

This systematic review was conducted based on the PRISMA extension for Diagnostic Test Accuracy (DTA) guidelines. A research protocol was prospectively prepared and registered. (PROSPERO: CRD42021241531, March 29, 2021).

Eligibility criteria

Inclusion criteria:

- Clinical trials, prospective and retrospective cohort studies, comparative studies, validation or evaluation studies, and epidemiologic studies.
- Medical and biological studies based on ~~the~~ an automated approach for detecting facial landmarks in 3D facial images.
- Studies that compare the accuracy of automated landmark algorithms ~~to~~ for manual identification and landmarking.
- Studies with clear outcome measures regarding the accuracy or reliability of 3D automated facial landmarking.
- Original articles published in English.

Exclusion criteria:

- Studies that include 3D facial landmarking for facial recognition (e.g., forensics~~,~~ and security), computer vision tasks, and other non-medically or biologically related purposes.
- Full text ~~is~~ not available or inaccessible.

Information sources and search:

Literature search strategies were developed using medical subject headings (MeSh) and keywords related to automated 3D facial landmarking. An electronic search of several databases including PubMed, EMBASE (OVID), Scopus, and Cochrane Central Register of Controlled Clinical Trials (CENTRAL) was performed in April 2021. There was no restriction ~~in~~ on the search period, but the search was limited to articles ~~in~~ on humans and published in English. Citations and cross-references from relevant key articles were manually searched to identify additional publications. This search was based on the PICO elements listed in Table 1. The search strategy is presented in Supplementary table 1.

Study Selection

Two reviewers (BA, AA), independently screened the titles and abstracts of the eligible studies using the Rayyan software (9). ~~They~~ The reviewers were not blinded to the authors' names, their affiliations, or the results of the eligible studies. The publication was considered relevant if one of the reviewers approved its inclusion.

Full texts of potentially relevant studies were retrieved and assessed by one reviewer (BA~~),~~) and then checked and approved by a second reviewer (AA). For each paper, inclusion criteria were assessed, and a final decision on eligibility was made. ~~The d~~Disagreements between the reviewers~~,~~ or areas where information was unclear, ~~was~~ were resolved by discussion. In cases where the same research group ~~have~~ had several articles on the refinement of an automated model, the latest publication was selected.

Data Extraction

One reviewer extracted the data from the selected studies and tabulated this information into a standardised form that was ~~checked~~ assessed by the second reviewer with respect to ~~Study~~ study design, ~~Imaging~~ imaging technique, ~~Sample~~sample, ~~Study~~ study methodology, ~~Algorithm~~ algorithm description, ~~Reference~~ reference ~~Method~~ method (manual landmarking), ~~Automated~~ automated algorithm evaluation, and outcomes.

Quality assessment

The two reviewers assessed the risk of bias and applicability of the included studies using a modified version of the Quality Assessment Tool for Diagnostic Accuracy Studies-2 (QUADAS-2) (10). Domains included patient selection, index test, reference standard, flow, and timing, with the first three domains also considered.

~~applicability~~ Applicability was rated on a 3-point scale: low (+), high (−), or unclear (?).

Synthesis of results

Given the considerable heterogeneity in ~~of~~ methodology ~~within~~ between the selected studies in terms of study design, sample, acquisition techniques, reference standards, and measured outcomes, a narrative synthesis was performed to identify and summarise the main findings of the included ~~studies~~articles.

Results

Study selection

Figure 1 presents ~~the~~ a PRISMA flow-chart for the article~~s~~ selection process in this study. From the 1002 manuscripts identified via the four databases, 549 studies remained after duplicate removal, 473 were excluded following title/abstract assessment and ~~following f~~full-text review of the remaining 76 articles, and 10 studies met the inclusion criteria. An additional 4 studies were found through a manual search ~~making~~ for a total of 14 studies that were eventually included in this systematic review. Among them, seven studies (11-17) were related to research question one~~1~~, five studies (7, 8, 18-20) were related to both research questions one~~1~~ and two~~2~~, and two studies (21, 22) were related to research question two~~2~~. A quantitative synthesis (meta-analysis) was not possible due to ~~their~~ the studies' methodological heterogeneity and different outcomes.

Study quality assessment

All ~~the~~ iincluded studies were critically assessed ~~by~~ using a tailored QUADAS-2 assessment tool (table 2). Regarding patient selection, twelve studies were rated as having a "high" or an "unclear" risk of bias (8, 11-21), and seven showed applicability concerns (8, 14-18, 21). This result was due to the absence of testing on consecutive or random subjects and an insufficient description of the population sample. Moreover, thirteen studies were found to be at an unclear or high risk of bias regarding implementing the reference standard due to the lack of an appropriate reference standard or ~~the~~ failure in reporting the results of inter- and intra-observer reproducibility assessments ~~(~~(6, 8, 11-17, 20-22).

Study characteristics:

The studies included in this review were published in English between 2012 and 2021, and most of them were based on retrospective studies ~~based on~~ using 3D facial images (~~table~~ Table 3). Supplementary Figure 1 shows the research topics of the included studies. The sources of 3D images used in the included studies varied. Two studies used data from a genetic cohort ~~(8, 19);~~ (8, 19), three studies used data from clinical settings ~~(7, 14, 22);~~ (7, 14, 22), one used the FaceBase Consortium ~~(11);~~ (11), one study used a dataset collected by the Collaborative Initiative on Fetal Alcohol Spectrum Disorders (CIFASD) ~~(12);~~ (12), one study used a dataset from the Prenatal Alcohol and SIDS and Stillbirth (PASS) Network in South Africa ~~(20)~~ (20), and six studies failed to report details on the source of the dataset (13, 15-18, 21).

Most studies omitted details of the sample characteristics (e. g., gender, age, inclusion and exclusion criteria, and main facial characteristics), one reported a prespecified sample size calculation (7), and two reported the use of randomization for population selection (7, 19), neither of which reported the randomisation details.

Table 4 shows the characteristics of the included studies from both test and reference perspectives. Different types of algorithms were used for automated 3D landmarking. The included studies were grouped according to the method used for the automated landmarking algorithm.

All ~~the~~ studies defined the reference standard (ground truth) by using the indirect anthropometry method, where landmarks and/or measurements were manually performed on 3D facial images by an observer, except for two studies (21, 22), where the ground truth was established by an experienced assessor taking ~~the~~ measurements directly ~~of~~ from the participants' faces ~~by an experienced assessor~~. Various numbers of facial landmarks were localised, ranging from 10 to 29 landmarks (Figure 2). The sample size of the training dataset ranged from 30 to 943 images and from 8 to 1000 images for the test dataset.

For interpretation purposes, the results were grouped based on the systematic review research questions~~:~~ (Q1 and Q2).

Description of individual studies

Research question one~~1~~: Accuracy of automatic 3D facial landmarking versus manual landmarking (n=9 studies)

For research question one~~1~~, landmarking accuracy was evaluated by comparing the mean distance between the automatically and manually detected landmarks (test versus reference standard). The overall mean errors and standard deviations were collected, when applicable, for all the ~~of the~~ included studies, which were grouped according to the method used for the landmarking, as follows.~~:~~

Atlas (template)~~-~~ based method:

A template with pre-defined landmarks is usually required for ~~template~~ template-based methods. Through ~~the~~ application of the estimated deformation field via registration, the landmarks for a testing image can be directly transferred from the corresponding landmarks in the template image. Two studies followed this method.~~:~~

Li et al. (8) developed an automated landmarking method to identify 29 facial landmarks by morphing ~~a 29- landmark~~a template face to an individual face through the guidance of anchor points. To validate ~~their~~ this method, a centroid size (CS) comparison was carried out to compare estimates of size across manual and automatic methods. CS estimates were found to be consistent between the two methods. The mean difference between the two methods was <1 mm, with a few landmarks showing mean differences of 1–2 mm.

Baksi et al. (7) combined template and texture-based methods to localise 21 facial landmarks. ~~Their~~ This approach required manual standardization of the reference position, and the average difference and standard deviation between the manual and automated methods for all landmarks was 3.20 +/- 1.64 mm. The points with a discrepancy of ≥2 mm were: Exocanthions, Glabella, Endocanthion (left), Alare (right and left), and Gnathion.

~~Knowledge~~ Knowledge-based method

This method is based on geometric information and prior knowledge of the typical shape of the craniofacial structure of the face and locates landmarks based on pre-~~agreed~~ established definitions.

Vezzetti et al. (17) proposed a new automatic method for localizing 13 facial landmarks based on geometric descriptors and a thresholding methodology to extract each landmark. The results showed a mean error of 4.7 mm, with the worst result obtained for subnasale, and the best result ~~was~~ achieved for ~~the~~ pronasale.

Abu et al. (18) used an automatic facial landmark model to locate 10 craniofacial landmarks ~~using~~ via information on ~~geometry~~ geometric characteristics ~~information~~.

The automated landmark locations were compared with the manually located landmarks, and the results showed that the average difference in location of the landmarks was 2.23 mm.

Combination and other approaches

Wang et al. (13) utilised visual facial pattern learning to identify each landmark's local geometric pattern and parameters utilising a small training set to localise 23 anthropological facial landmarks. A randomly selected 3D face, together with its manually placed landmarks, was chosen as a template face, to align all other 3D faces and g~~to g~~enerate the positions of the initial landmarks. Then, a Visual Analytical Approach algorithm was employed to refine the projected landmarks. The overall mean error of the automated landmarking method was 2.23 mm with midline points exhibiting less errors.

Guo et al. (15) predicted 17 landmarks using a principal component analysis (PCA)~~)~~)-based method to localise 6 landmarks on the sample surfaces. Then, 11 additional landmarks ~~are~~ were identified heuristically using geometric and texture relations. The average root mean square error of the manual landmark localisation method was ~1.7 mm, with most SD values below 1mm.

Liang et al. (16) developed a combined geometric and template method to automatically locate 20 facial landmarks. Compared to landmarks manually identified by medical experts, the results showed an average difference of 2.64 mm.

De Jong et al. (6) used the Elastic Bunch Graph Matching-based (EBGM) algorithm to identify 21 facial landmarks. Using cross-validations (leave-one-out experiments), ~~they~~ the authors reported ~~an~~ average accuracy of 1.7 mm, which showed a 22% improvement over their previous algorithm ~~-~~ (23).

Sukno et al. (14) used a regression algorithm to generate candidate points from geometric cues extracted ~~by~~ using asymmetric~~y~~ pattern ~~s~~ shape context~~s~~ (APSC) descriptors. The localisation accuracy was evaluated on a clinical dataset of 144 facial scans, and the results showed an average error of 2.3 mm, with errors reported to be below 2 mm for more than half of the targeted landmarks.

Deep learning-based model

~~When c~~Compared ~~with the~~ to more classical methods of automated landmarking, deep learning-based methods are learnt hierarchically through the multiple layers of

an artificial neural network (ANN). Convolutional Neural Networks (CNNs) are by far the most widely used type of ANN.

Bannister et al. (11) applied a pretrained automatic 3D facial landmarking algorithm on a test sample of 444 facial images based on an ensemble of regression trees cascade trained on large databases of normal facial images .(24). Ground Truth landmarks were localized by two individuals, and the average distance between the positions of manually located landmarks and those automatically generated was 2.5 mm. The largest errors were reported at the eye landmarks (en, ex), the nasion landmark (n), and the gnathion landmark (gn).

Fu et al. (12) proposed utilising limited training samples through a cross-task representation learning approach to produce a deep learning framework for anatomical landmark detection. The results showed ~~the~~ that effectiveness of the proposed approach~~ has~~ outperformed other~~ compared~~ solutions with a mean error and standard deviation value of 0.83~~0~~ ± 0.25~~45~~, respectively.

Huang et al. (20) used a pre-trained CNN model that integrated a novel layer to extract landmark~~s~~ coordinates from 3D facial images. The~~ir~~ dataset was randomly assigned to training (n = 622) and test (n = 155) sets. ~~They~~ An expert manually annotated 20 reliable landmarks on each 3D facial image~~ by an expert.~~, and the landmarks were then manually projected to each corresponding 2D texture map as training labels for the CNN models. The localisation accuracy was assessed by measuring the Euclidean distance between the gold standard reference and the predictions in 2D. The mean error was 0.67 mm (standard deviation 0.63 mm).

Research question two~~2~~: Reliability of further analysis obtained through automated landmarking versus manual landmarking (n= 6 studies)

Research question two~~2~~ assessed the reliability of measurements and ~~analysis~~ analyses using~~obtained by~~ automated landmarking compared to manual landmarking. The type of measurements and ~~analysis~~ analyses varied according to the research area of the included study, either genetic or clinical.

Genetic studies

Two studies evaluated the performance of automated landmarking based on analysis that is relevant to genetic and developmental studies.

In Li et al. (8), both manual and automated landmarking of 79 scans ~~were~~ was performed,~~ and a comparison of~~ and variation in shape structure was ~~carried~~ compared out using PLS regression. It was found that the two methods yielded

variance–covariance matrices that were highly correlated (P <0.001 based on 1000 permutations); thus, automated data produced covariation structures that closely resembled manual landmarks.

De Jong et al. (6) used heritability as a benchmark to evaluate landmarking performance and found that the estimated heritability was improved compared to ~~their~~ the authors' previous algorithm~~, which was~~ based on 2D Gabor wavelets (23). Twin-based heritability information was estimated for several features derived from landmark coordinates. The results showed that distances were the most heritable traits and were concentrated in the mid-face region.

Clinical studies

Five studies used linear facial anthropometric measurements ~~that are~~ relevant for clinical use to evaluate the performance of their automated landmarking method.

Huang et al. (20) ~~r~~eported the accuracy of ~~the~~ automatically generated palpebral fissure length (PFL) measurements ~~,~~ derived from 3D facial images, which could be used ~~in assisting~~ to assist the diagnosis of ~~fetal~~foetal alcohol spectrum disorder.

The~~ir~~ model showed a mean error of 1.2 ~~mm~~ ± 0.8 mm~~,~~ when compared to ~~the~~ manual annotation~~, which~~ and was ~~the lowest in comparison to~~lower than two different CNN models (regression-based and synthetic heatmap matching models).

~~-~~**Lippold et al. (22)** reported ~~a~~ substantial agreement between digital and manual measurements for 4 out of 7 evaluated distances, the. f~~The f~~ormer being~~:~~ intercanthal distance, the distance between outer eye corners, the subnasale distance ~~subnasale~~ to the mentolabial point, and subnasale to soft tissue pogonion.

Baksi et al. (7) found ~~a~~ moderate to good agreement (ICC > 0.5) between ~~the~~ manual and automated landmark identification processes in 5 out of 10 linear measurements, which only included ~~only which were:~~ nasal width, labial fissure, upper vermilion, and upper and lower facial height.

Abu et al. (18) evaluated 8 linear measurements extracted by both automated and manual methods. The ~~and the~~ average difference was 2.16 ~~mm~~ ± 2.25 mm.

Palmer et al. (21) incorporated the MeshMonk (25) tool-box into ~~a platform,~~the Cliniface software platform that was~~that has been~~ designed to support facial anthropometric analysis. Ten linear facial measurements were compared against the expert's manual measurements, and 3 out of 4 were found t~~o be at a level of~~o provide a level of accuracy comparable to that of the manual expert.

Accuracy of individual landmarks

The mean errors and standard deviations (where applicable) for the 23 most commonly ~~-~~reported landmarks were collected from 9 studies. The ~~and the~~ results are ~~is~~ presented in table 5.

Discussion

The present systematic review evaluated the performance of automated landmarking on 3D facial images by ~~looking int~~examining~~o~~ two different metrics: ground truth-based localisation errors (Research Question one) and task-oriented performance (Research Question two). For ground truth-based localisation errors, ~~a~~ manually annotated ground truth (reference standard) was used to assess ~~the~~ landmarking performance. For task-oriented performance, we ~~looked into~~ explored studies that measure the impact of landmarking accuracy on the performance score of ~~a~~ the task.

Accuracy of the automated model

Concerning research question one, the best localisation results were found in the two studies that used a ~~deep~~ deep-learning learning-based 3D landmark detection method ~~-~~(12, 20). Although these results ~~could be argued to~~ indeed supported the claim that automated algorithms can match "ground truth" accuracy, ~~however~~ several methodological ~~-~~ shortcomings were common in most of the included studies and should be considered before accepting these results.

The majority of the included studies showed a range of risks of bias and applicability concerns. Data selection produced small and possibly non-representative populations. Additionally, t~~T~~he training and test sets ~~was~~ were not clear in most of the studies, and precautionary measurements to avoid overlap between the groups were not mentioned. However, inadequate reporting in a study does not automatically mean that the study was poorly designed, and a poor study design does not necessarily indicate a poor quality automated landmarking algorithm. Nonetheless, both of these deficiencies affected the comparison between studies and whether or not meta-analyses ~~are~~ were possible.

Reference standard

Validating the accuracy of the automatic landmarking algorithms against expert manual landmarking (reference standard) is essential as i validationt ~~would~~ provides

a more objective measurement tool. ~~Also,~~Additionally, understanding the error and variation introduced by manual landmarking is important as some learning learning-based automated landmarking approaches are influenced by manual annotation errors, and ~~their~~ performance may vary depending on the annotator. ~~Yet~~ However, most of the studies showed a high risk of bias regarding the implementation of the reference standard due to inadequate detail ~~on~~ in the assessment, interpretation, and reproducibility~~y, and~~y. This result ~~this~~ i is in line with other recent systematic reviews on automated cephalometric landmarking (26, 27).

Therefore, extensive ~~quality~~ quality-control procedures for establishing the ground truth in future automated 3D landmarking studies should include ~~a~~ clear definitions of the tested landmarks, understanding ~~by~~ for each annotator, and training plus calibration before engaging in landmarking. It ~~has been~~was also suggested that every study should include an assessment of the inter- and ~~intra~~ intra-observer reproducibility of manual landmarking even ~~where~~ when standard measurements are used (28).

Type of landmarking techniques

The studies included in this review used various approaches for automatic landmarking, including knowledge, atlas, learning, and ~~hybrid~~ hybrid~~based~~ methods, each of which has its own limitations. The results of the classical learning approach ~~is~~ are always dependent on the training data, and errors increase if the variability of testing data is comparatively higher than that of training data. Knowledge-based approaches involve the~~s~~ detection of geometric features, typically associated with local curvature~~,~~ that match the shape properties of the landmark. However, this method cannot detect all ~~kind~~ types of landmarks, especially those ~~which are~~ not located on any significant edge. The main problem of ~~atlas~~ atlas-based methods is that they largely rely their reliance on the accuracy of cross-subject registration. Although multi-atlas registration can improve ~~the~~ robustness and accuracy, it is very time intensive to perform multiple registrations for each testing subject. The majority of studies used combined approaches by considering 3D geometry of the face surface and the associated texture images. Deep learning-based methods ~~have shown~~ were found to be superior~~ity~~ in identifying anatomical landmarks in medical images, ~~and this~~ which was also supported in ~~the~~ this review, but only ~~3~~ three studies followed this approach (11, 12, 20). This result is, perhaps, not surprising ~~,~~ since the deep learning approach usually requires a large number of~~of~~

~~training~~ 3D facial images to be trained with manually annotated landmarks ~~and given that~~ . Additionally, expert landmarking is often expensive and time-consuming, ~~therefore~~ making it difficult to generate large high-quality datasets suitable to train such models.

Many 3D data acquisition technologies enabled the direct collection of 3D surface images of human faces, along with their high-resolution matching 2D texture images. Literature resources are more abundant in studies carried out for detecting landmarks in 2D images, given that 3D imaging is more recent. Consequently, many studies in this review utilised 2D landmarking algorithms to detect landmarks in 3D ~~by~~ using projection transformation (12, 15, 19, 20). Once detected, the 2D landmarks can be projected back onto the 3D image. The advantage of using 2D data of 3D images, rather than using the 3D images directly, is reducing the computational cost of processing of 3D data. This computational cost is high and also requires a large amount of memory. By contrast, the 2D data can alleviate the memory limitations since the landmarking is conducted in a set of 2D images.

However, 2D algorithm is known to be sensitive to pose variation and facial expression which could lead to poor localization performance with this technique. This can be avoided by following a standardized methods of data acquisition and collection that is usually followed in the clinical and biological settings.

Selection of landmarks and Individual landmark errors

The number of landmarks in this review ranged from 10 to 29, depending on the application of the intended analysis. The most commonly selected landmarks captured the facial soft tissue units (eyes, nose, and mouth). Table 5 shows the 23 ~~most~~ most commonly reported landmarks and their localisation errors.

The magnitude of error for ~~the~~ individual landmarks varied among the included studies ~~. This is~~ due to the heterogeneity in the methods used to detect the same landmark. There was ~~a~~ general agreement that midline landmarks are more accurate when localised automatically compared to laterally localised landmarks. Gnathion and Glabella were difficult to automatically identify in some of the included studies, which could be due to the anatomical differences between the subjects. This pattern of errors is similar to that found with manual landmarking in previous studies .(29-31).

Although endocanthion and exocanthion (inner and outer eye corners) are considered ~~as~~ to be reliable landmarks in manual landmarking techniques, as reported by previous studies (32, 33), some of the included studies found ~~those~~ these landmarks among the most difficult points to ~~be located by t~~ locate using the automated method ~~.~~ (11, 14, 16). Bannister et al. (11) visually ~~checked~~ assessed images with the highest errors in the automated landmarking group and found that closed eyes were common among those groups. Guo et al. (15) found pogonion and earlobes to be among the worst landmarks ~~to be~~ located automatically, and those landmarks were highly influenced by the presence of facial and head hair, which could contribute to the relatively larger errors and standard deviations. It ~~has been~~was suggested previously that any loose hair should be pulled away from the face and ears ~~during~~ when capturing of the face, if these landmarks are to be used in the analysis (34). This result emphasised the importance of quality control in the data acquisition phase to ensure the quality of the captured 3D images.

Anatomical facial landmarks are essential for various clinical applications like such as image analysis and diagnosis ~~that~~ , which support the clinical decision-making process. This factor was reflected in the present~~is~~ review~~,~~, as most of the studies were based on clinical or medical settings. ~~Yet~~ However, only three~~3~~ studies included a clinical population cohort ~~for~~ in their datasets (7, 12, 20). Having representative data from the clinical population is essential, especially in learning algorithms, as ~~otherwise,~~ selection bias is likely otherwise (35).

In clinical use, landmark accuracy is important, as there is a need to detect subtle shape differences or growth changes. Depending on the research group, localisation and reproducibility errors are considered clinically relevant when these~~y~~ exceed 0.5 ~~mm~~(31), 1 ~~mm~~(30) or 2 mm (36). Hajeer et al. (31) considered a landmark to be highly reproducible when the standard deviation (SD) from the centroid was 0.5 mm or less. Toma et al. (37) set a reproducibility threshold of less than a 1 mm difference ~~the~~ for the landmarks to be clinically acceptable. The clinical significance of ~~the extent of~~ landmark identification errors relies on the level of accuracy required. If the 3D images are used for surgical planning, sufficient accuracy is needed to allow the clinician to plan treatment accurately and evaluate surgical outcomes. In this review, most studies did not report a cut-off value for acceptable errors ~~for~~ in clinical and biological use. Table 5 Shows the landmark identification errors for 23 of the most commonly reported landmarks in the included studies. Assuming that a mean error within 2 mm is acceptable, 68% of the estimated landmarks in the included studies ~~will fall~~fall within an~~the~~ acceptable

range. By reducing the acceptable mean error to ~~be~~ within 1 ~~mm~~ and 0.5 mm, only 5% and 0% of the detected landmarks, respectively, ~~will fall~~ fall within ~~the~~ an acceptable range~~, respectively.~~ (Supplementary tables 2,3,4).

Compared ~~with~~ to the number of published articles on the development of automated landmarking ~~on~~ in 3D cephalometrics, the development of automated landmarking in 3D facial images is lagging considerably ~~lagging~~ and requires greater attention.

Limitations of the current literature and future directions (Research question three)

This review found that, with the current error levels reported in the literature~~,~~ the automatic landmark localisation of individual landmarks is not sufficiently accurate ~~enough~~ to allow ~~their~~ the use of this technique for clinical purposes~~. This is~~ , as~~because~~ the errors in landmark~~s~~ detection ~~were~~ remain higher than those obtained ~~with~~ via manual landmarking. This result indicates that automatic facial landmarking is still developing, and further studies are required to create~~develop~~ a system that ~~could~~ can match or outperform the accuracy of the current gold standard. ~~this~~ This systematic review was limited by the quality of the included studies and uncovered several methodological limitations evident in ~~this~~ the corresponding literature. The most significant source of bias found was related to the potentially selective and incomplete reporting of results. This factor could have impacted the outcomes of the included studies and ~~could~~ lead to overfitting of the tested algorithms. Other key issues ~~of~~ related to inadequate study design and reporting were found within the included studies ~~in~~ when implementing ~~the~~ reference standards and population selection ~~for the included studies~~. Evaluation of the performance metrics of the included studies was also limited by the variation of the statistical methods applied. (supplementary table 5,6).

A common limitation of the current automated approaches in the included studies was ~~having a~~manually carrying out the pre-processing stage ~~of~~ for 3D facial images ~~carried out manually~~. This process include~~d~~s placing the image in the correct pose and orientation or cropping the face by removing hair, clothing, and background ~~in~~ from the image before locating the tested landmarks (7, 8). Advancements in pre-processing steps are highly recommended~~, and f~~. Future studies should aim to avoid the manual pre-processing step and develop models that can localize facial landmarks regardless of their orientation.

This systematic review identified some areas for future research. Large studies with adequate power are essential for achieving a more robust and reliable evaluation of the automated landmarking algorithm. Convolutional Neural Networks (CNNs) have recently emerged as the mainstream approach for facial landmarking and have demonstrated superior performance in medical tasks (38-40). By utilising the CNN approach, significantly less time-intensive and ~~more adequate~~more accurate ~~accuracy of~~ automated 3D facial landmarking in clinical settings~~ could be~~is expected compared to manual landmarking. However, researchers should remember that when using deep learning models, the high-performance results of any newly developed model can only be accepted in~~ the~~ research and clinical communities through well-designed and well-conducted studies that reduce bias and report results thoroughly and transparently.

Conclusion

This systematic review ~~has~~ highlighted a few crucially important points as we aspire to achieve automated landmark annotation in 3-D facial imaging.

1- The overall performance of~~ the~~ automated 3D facial landmarking models ~~varies~~ varied ~~among~~ between the different algorithms used, and the best performance was achieved~~ by~~ using deep learning models. ~~Yet~~ However, it is still necessary to verify the generalisability and reliability of these models by using adequate images for a representative population and conducting a well-designed study that minimises bias and is transparently reported.

2- Automated landmarking models of 3D facial images in clinical and biological settings at present are unable to compete with manual landmarking in terms of the accuracy of landmark detection.

3- Future studies should focus on implementing the recent advancements of the deep learning models in developing automated 3D facial landmarking for clinical and biological purposes. Researchers should bear in mind that to be robust, deep learning techniques are heavily influenced by the quality and quantity of the training data.

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Ethics approval information

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Data availability

The data underlying this article are available in the article and in its online supplementary material

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Figures and Tables captions

Figure 1: PRISMA Flow chart illustrating the study selection process.

Figure 2: The 3D facial landmarks that have been used in the included studies. **Midline landmarks**: g=glabella, n=nasion, prn=pronasale, sn=subnasale, ls=labialesuperius, st=stomion, li=labialeinferius, sl=sublabiale, po=pogonion, g=gnathion, me=menton. **Bilateral landmarks**: en=endocanthion, ex=xocanthion, mso= mid-supraorbital, mio=mid-infraorbital, da=dacryon, al=alare, ac= alar curvature, sag=superior alar groove, sa=subalare, ect=ectoconchion, ss=subspinale, cp= crista philtri, ch=chelion, sl=sublabiale z= zygion. (Tragion and earlobe tip landmarks are not illustrated as the ears area is not shown in this 3D model).

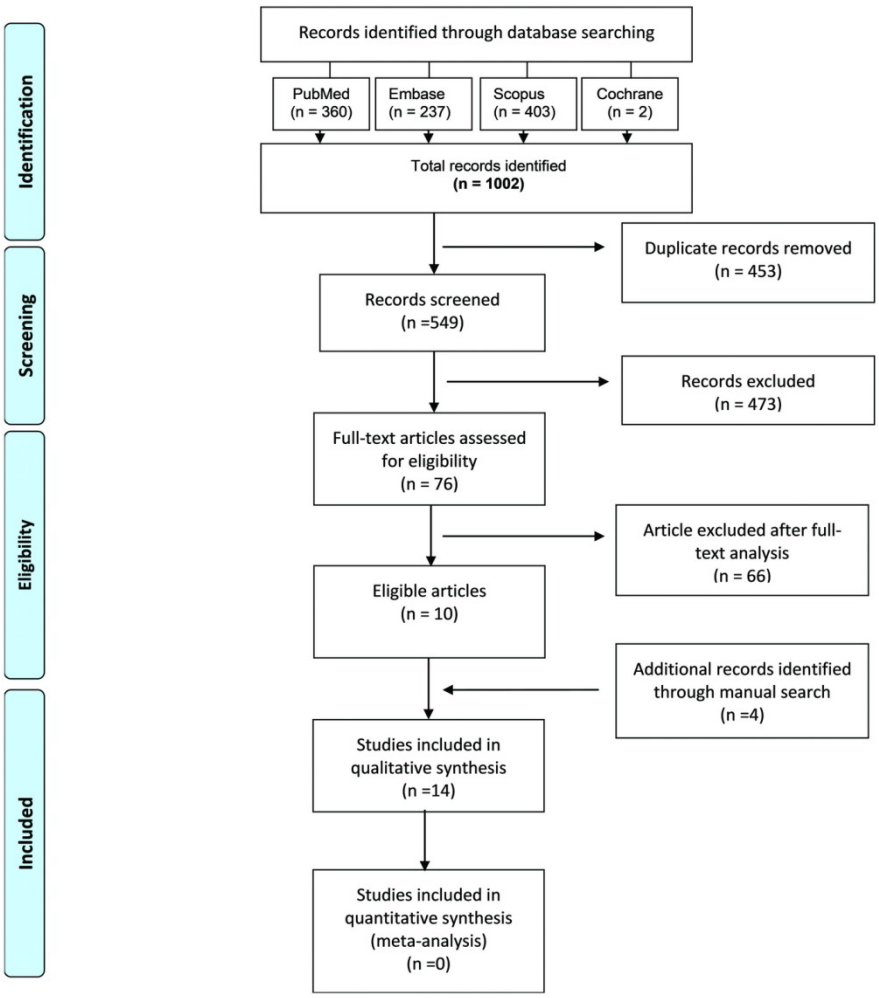
Table 1: Description of PICO (P=Population, I=Intervention, C=Comparison, O=Outcome) elements and keywords.

Table 2: Bias and applicability assessment of included studies using tailored QUADAS-2 tool.

Table 3: Overview of the characteristics of all the included studies.

Table 4: Characteristic of the included studies from test and reference perspectives.

Table 5: Landmark identification error for 23 of the most reported landmarks in the included studies.



134x157mm (300 x 300 DPI)

Criteria	*Specification*
Focus questions	Q1: "What is the accuracy of automatic 3D facial landmarking when compared to manual landmarking. And which of facial landmarks are more or less reliable.?" Q2: "What is the reliability of the soft tissue facial analysis or measurements obtained through automatic landmarking when compared to manual landmarking"
Population	3D facial images of the general or disease specific human population, with no age limit;
Intervention	Automated landmarking of 3D human facial images;
Comparator	For Q1: - Automatic 3D landmarking versus the reference: manual landmarking (ground truth); For Q 2: - 3D facial analysis/ measurements obtained through automatic landmarking versus those obtained through manual landmarking;
Outcome	To assess the performance of the 3D automated landmarking technique; For Q1: - Mean differences and standard deviations expressed in mm (Euclidean method) between the tested method and the reference; For Q 2: - Mean differences and standard deviations (expressed in mm) for facial analysis between the tested method and the reference.
Keywords	(((((((Landmarking) OR (Landmarks)) OR (Anatomic landmarks[MeSH Terms])) AND (((Face) OR (Facial)) OR ("Face/diagnostic imaging"[Mesh]))) AND ((((("Imaging, Three-Dimensional"[Mesh]) OR (three-dimensional)) OR (three dimensional)) OR (3D))) AND (((((((((((Valid*) OR (Accuracy)) OR (Accurate)) OR (Repeatability)) OR (Agreement)) OR (Concordance)) OR (Reproducibility)) OR (Reliability)) OR (Compar*)) OR (Performance)) OR (Error)) OR ("Reproducibility of Results"[Mesh])) OR ("Observer Variation"[Mesh]))) AND (((((Automat*) OR (Detection)) OR (Identification)) OR (Localization)) OR (Algorithm))) AND ((((("Humans"[Mesh]) OR (Human*)) OR (Male*)) OR (Female*))

Table 1: Description of PICO (P=Population, I=Intervention, C=Comparison, O=Outcome) elements and keywords.

114x147mm (300 x 300 DPI)

Study	Risk of Bias				Applicability Concerns		
	Patient Selection	Index Test	Reference Standard	Flow and Timing	Patient Selection	Index Test	Reference Standard
Baksi et al 2020	☺	☺	☺	☺	☺	☺	☺
Bannister et al. 2020	?	☺	☹	☺	☺	☺	☺
Fu et al 2020	☹	☺	☹	☺	☺	☺	?
Palmer et al 2020	☹	☺	?	☺	☹	☺	☺
Wang et al. 2019	☹	☺	☹	☺	☺	☺	☺
Abu et al 2019	☹	☺	?	☺	?	☺	☺
Huang et al 2019	☹	☺	☹	☺	☺	☺	☺
De Jong et al 2018	?	☺	☹	☺	☺	☺	☹
Vezzetti et al. 2018	☹	☺	☹	☺	☹	☺	☹
Li et al. 2017	?	☺	?	☺	?	☺	☺
Sukno et al. 2015	☹	☺	☹	☺	?	?	?
Lippold et al 2014	☺	☺	☹	☺	☺	☺	☹
Guo et al 2013	☹	☺	☹	?	?	☺	☹
Liang et al 2013	☹	?	☹	☺	☹	☺	☹

☺ Low Risk ☹ High Risk ? Unclear Risk

Table 2: Bias and applicability assessment of included studies using tailored QUADAS-2 tool.
143x112mm (300 x 300 DPI)

Author/year	Research Area	Study Design	Population Demographic and Selection Method									
			Sample Source	Sample Number	Sample Size Calculation/ Justification	Age	Sex	Ethnicity	Random or Consecutive Inclusion	Inclusion Criteria	Exclusion Criteria	Facial Characteristic of Subjects
Baksi et al 2021	Ortho	Retro	Orthodontic patient	30	R	9-17 y	15 M 15 F	Caucasian	Randomly selected.	Full 3D facial image without artefact or distortion.	craniofacial anom-alies or significant asymmetry	NR
Bannister et al. 2020	Facial Syndrome	Retro	FaceBase Consortium	444	NR	under 1 y - 75 y	NR	NR	NR	NR	NR	Syndrome associated with facial dysmorphology.
Fu et al 2020	Medical (Fetal Alcohol Spectrum Disorders)	NR	Collaborative Initiative on Fetal Alcohol Spectrum Disorders (CIFASD)	1549	NR	NR	NR	NR	NR	NR	NR	NR
Palmer et al 2020	Clinical/ biological	Pro	NR	50 3D image of 25 subjects	NR	NR	NR	NR	NR	NR	NR	NR
Abu et al 2019	Ortho	NR	Unclear	60	NR	18-25 y	30F 30 M	NR	NR	NR	NR	NR
Huang et al 2019	Medical (Fetal Alcohol Spectrum Disorders	NR	Prenatal Alcohol and SIDS and Stillbirth (PASS) Network in Stellenbosch, South Africa	772	NR	1-2 months	NR	Mixed (not specified)	NR	3D images of unexposed and prenatal alcohol-exposed infants at 1–12 months	NR	NR
Wang et al. 2019	Genetics	NR	Unclear	39	NR	NR	nearly equal sex ratio	African/ Asian/ European/ South American/Middle Eastern ancestry	NR	NR	NR	NR

Table 3: Overview of the characteristics of all the included studies.

151x99mm (300 x 300 DPI)

Author/year	Research Area	Study Design	Population Demographic and Selection Method									
			Sample Source	Sample Number	Sample Size Calculation/ Justification	Age	Sex	Ethnicity	Random or Consecutive Inclusion	Inclusion Criteria	Exclusion Criteria	Facial Characteristic of Subjects
De Jong et al 2018	Genetic	Retro	The Twins UK cohort.	40	NR	NR	NR	European descent	Randomly selected	NR	NR	NR
Vezzetti et al 2018	Medical	NR	NR	230	NR	NR	NR	NR	NR	NR	NR	NR
Li et al. 2017	Genetic	NR	GWAS of facial shape in Tanzanian children (Cole et al. 2016)	79	NR	NR	NR	Tanzanian children	NR	NR	NR	NR
Sukno et al. 2015	Clinical	NR	Training set: FRGC Public dataset Clinical set: Healthy subjects/ source: NR	943 + 144	NR	NR	NR	NR	NR	NR	NR	NR
Lippold et al 201d	Ortho	Pilot pro	Orthodontic residents and employees of dental clinic	15	Pilot study	21.8 - 40.6 Mean: 28.2 SD: 5.5 y	10 F 5 M	NR	NR	NR	NR	NR
Guo et al 2013	Genetic/ Medical	NR	NR	1080	NR	18-28Y	R	Uyghurs Chinese / Han Chinese	NR	NR	NR	NR
Liang et al 2013	Medical	NR	NR	115	NR	NR	NR	NR	NR	NR	NR	NR

Ortho=Orthodontics, Retro=Retrospective, R= Reported, NR=Not reported, Pro=Prospective

Q1 studies / Q2 studies / Q1+Q2 studies

Table 3: Overview of the characteristics of all the included studies.

145x87mm (300 x 300 DPI)

Author / Year	3D AS	No of LM	**Index Test – Automatic 3D Facial Landmarking**			**Reference Standard – Manual Annotation**				The Performance Evaluation Method	**Main Results:** Total Mean Difference and SD Between Index Test and Reference Standard	**Landmarks with Good Localization**	**Landmarks with Poor Localization**
			Algorithm Used	Training Dataset	Test dataset	Number of Observers	Observer's Experience Level	Observers Repetition	Intra-Observer /Inter-Observer Results				
Baksi et al 2021	3dMD	21	Template based	N/A	30	2 observers	Experienced orthodontist + orthodontic resident	observer 1 /2 weeks interval/for all 30 facial scans	Intra-class correlation coefficient for intra-observer reliability and inter-observer variability was >0.9	Direct comparison of manual vs automatic landmark placements.	M ± SD 3.2 ± 1.64 mm. 16 out of 21 landmarks had a mean difference < 4 mm.	The points with the greatest agreement (≤2 mm) were mainly located in the midline: prn, sn, ss, ls	Glabella showed the largest discrepancy. LM with discrepancy ≥4 mm were ex (right), al (right), sl (right), and gn.
										10 linear measurements	Only five had a moderate to good agreement (ICC > 0.5) between manual vs automated identification processes		
Bannister et al. 2020	3dMD	12	Deep convolutional regression network (pretrained algorithm)	N/A	444	2 observers (no specification of their background)	NR	2 times for one observer (subset of 100 sans) One day interval	Slightly lower overall average intra-observer landmark distance than that of inter-observer distance with values of 1.1 and 1.5 mm, respectively	Direct comparison of manual, automatic and semi-automatic (NICP) landmark placements.	Automatic algorithm= 2.5 mm NICP= 3.1 mm	Mouth landmarks (ch, ls, li), nose landmarks (prn, sn)	eye landmark(en,ex), n, gn

Fu et al 2020	3dMD	NR	Cross-task representation learning (dep learning)	80%	20%	1 (expert)	Expert	NR	NR	Mean error + Failure Rate (FR) > 1.2+ Area-Under-the-Curve (AUC)	M ±SD: 0.830 ± 0.245 FR: 7.74% AUC: 0.32	NR	NR
Palmer et al 2020	3dMD + Vectra H1 (handheld)	N/A	Template based	N/A	50	1 Expert (direct manual measurement on participants face) + 4 non expert (indirect measurements)	Experienced clinical geneticist	NR	NR	Direct comparison of linear measurements from manual (ground truth), indirect (non-expert assessor) and automated (cliniface) methods	RMSE of Cliniface's measurements was 4.87 mm.	N/A	N/A
Abu et al 2019	Vectra -M5 360	10	Geometric based	N/A	60	1	NR	NR	The ICC test was performed to check the reliability of the linear measurements data taken manually. ICC is 0.99, with a 95% confidence interval, CI (0.98, 0.99)	Direct comparison of manual and automatic landmark placements.	2.23 mm	Ch and A.	Sn, sto, ls and li
										8 linear measurements and 5 proportional indices extracted from the nose and orolabial region	2.16 mm ± 2.71		

Huang et al 2019	3dMD	20	CNN model	622	155	1 (expert)	Expert	NR	NR	The localization accuracy was evaluated by calculating the Euclidean distance between the ground truth (GT) and the predictions in 2D	0.67 mm ± 0.63	NR	NR
										Normalization mean error 3D	4.3 ± 3.9		
										palpebral fissure length (PFL) error	1.2 ± 0.8		
Wang et al 2019	Vectra H1 handheld	23	Combined approach (template based + visual analytics approach) (VAA)	Training set:30, one image was used as template face	8	2 observers	expert+ student	I time	NR	Direct comparison of manual and automatic landmark placements.	2.23 mm	the midline LM, tend to exhibit less error than the bi-lateral LM.	mio, z, go (R), Ect (L) and, n
De Jong et al 2018	3dMD	21	Combined approach (Statistical ensemble)	40	Used the same set of trainin	1 the ground truth consisted of a single	NR	Unclear	NR	cross-validation (leave-one-out experiments)	1.7 mm +/- 0.4	Landmark 7 and 13 (Nose landmarks)	Landmark 9. (upper eyelid)

					g set (cross validation)	manual labelling of the entire data set.				Heritability estimate	The highest observed heritability was 87% for the area of the triangle defined by landmarks 10, 12, and 18. The best distance had heritability of 72%. Angles and coordinates had best heritabilities of 69% and 64%, respectively. Heritabilities improved by ~5%		
Vezzetti et al 2018	NR	13	Geometric based	N/A	230	NR	NR	NR	NR	Direct comparison of manual and automatic landmark placements.	4.73 mm	prn	The worst results were obtained for sn
Li et al 2017	Creaform Megacapturor and Gemini 3D camera systems.	29	Template based	50 (to create average face as a template)	Exp1: 30	2 observers	NR	4 times. Different days (interval not specified)	MANOVA showed that observer error accounted for (7%) despite significant attempts to standardize landmark placement. Measurement and inter-	Centroid size (CS) comparison to estimates of size across manual and automatic methods	CS estimates were consistent between the two methods (r =0.97, P <0.0001). The differences between the manual and automated methods average <1 mm, with a few landmarks showing mean differences of 1–2 mm	Sn, li, and ls	Tr and Z

					Exp2: 79				observer error for manual landmarking differ among landmarks. The results demonstrated that manual landmarking, particularly by multiple observers, was associated with significant measurement error.	Overall shape variation (multivariate analysis of variance-MANOVA)	The PLS analysis showed that the two methods yield variance–covariance matrices that are highly correlated. Landmarking errors occurred in about 5% of the Tanzanian sample		
Sukno et al. 2015	Handheld laser scanner	14	Asymmetry pattern and shape regression based	943	Clinical data set: 144	NR	Expert	NR	NR	Direct comparison of manual and automatic landmark placements.	overall accuracy of 2.3 mm, with typical errors below 2 mm for more than half of the targeted landmarks	N, en and sn	Po and ex
Lippold et al 2014	FastSCAN ™ laser-scanning system	NR	Geometric based (Cervature map and profile analysis)	N/A	15	1	experienced orthodontist	3 times for one image by same observer.	The three facial scans of a single patient on three consecutive days yielded almost identical values for all seven distances evaluated in	11 linear facial measurements	Lin's CCC showed substantial agreement between digital and manual measurements for 4 of the 7 distances evaluated.	N/A	N/A

									this study.				
Guo et al 2013	3dMD	17	Combined approach (geometric and texture based)	80 Han Chinese	Landmarking accuracy: 239=115 Han Chinese+ 124 Uyghur Chinese.	1	NR	NR	NR	Direct comparison of manual and automatic landmark placements.	mean errors between 1mm and 1.5mm, indicating rather high accuracy. Most of the SD values are below 1mm, suggesting good consistency of the annotation across different samples.	prn and ch	Po and earlobe tips
					Robustness of registration:1000=684 Han Chinese + 316 Uyghurs Chinese					The mean and standard deviation (SD) of the annotation errors measured in Euclidean distance and root mean square (RMS) errors were calculated	RMS error is within the mean range 1.1-2mm		
										Robustness of the registration method	Most parts have point-wise errors much less than 0.9 mm, indicating robustness of registration method		

											with varying references.		
Liang et al 2013	3dMD	20	Geometric and Template based	N/A (only one template mesh used)	115	1	Medical expert	NR	NR	Direct comparison of manual and automatic landmark placements.	2.64 mm	Prn and sa	Gn and ex

NR=Not reported; N/A=Not applicable, AS= acquisition system, R= right, L= left, g= glabella, ex= exocanthion, en=endocanthion, n=nasion, prn=pronasale, sn=subnasale, al=alare, sa=subalare, ss=subspinale, ls=labiale superius, li= Labiale inferius, ch= chelion, sto=stomion, sl=sublabiale gn=gnathion, ,go=gonion, z=zygion, mio=mid-infraorbital, tr=tragion ect=ectoconchion,

Q1 studies Q2 studies Q1+Q2 studies.

		Guo et al 2013	Liang et al 2013	Sukno et al 2015 (clinical dataset group)	De Jong et al 2018	Vezzetti et al 2018	Abu et al 2019	Wang et al 2019	Baksi et al 2020	Bannister et al 2020	Min	Max
	No of LM	**17**	**20**	**14**	**21**	**13**	**10**	**23**	**21**	**12**		
1	**Nasion**	1.49	2.92	1.72	2.00	3.05	1.44	6.17	3.05	3.10	1.44	6.17
2	**Pronasale**	0.76	1.59	1.90	1.40	2.78	1.22		2.00	1.90	0.76	2.78
3	**Subnasale**	1.29	2.45	1.83	1.50	6.54	5.79	2.52	1.65	2.40	1.29	6.54
4	**Labiale superius**	1.27	2.27	2.19	1.50		3.49	3.84	1.51	2.50	1.27	3.84
5	**Stomion**	1.19	1.49				3.51		1.84		1.19	3.51
6	**Labiale inferius**	1.38	2.27	2.35	2.10		3.23	1.52	2.35	2.10	1.38	3.23
7	**Endocanthion (right)**	1.19	2.39	1.73	1.60	5.02			3.13	3.40	1.19	5.02
8	**Endocanthion (left)**	1.15	2.78	1.73	1.60				3.80	2.90	1.15	3.80
9	**Exocanthion (right)**	1.34	3.34	3.21	1.80	6.16			3.44	2.50	1.34	6.16
10	**Exocanthion (left)**	1.16	3.68	3.21	1.80				4.45	2.00	1.16	4.45
11	**Alare(R)**	1.31	1.78			3.57	−0.83		4.20		1.31	4.20
12	**Alare(L)**	1.48	3.07						3.44		1.48	3.44
13	**Alar curvature point (right)**			2.01	1.30			2.41			1.30	2.41

Table 5: Landmark identification error for 23 of the most reported landmarks in the included studies.

155x105mm (300 x 300 DPI)

		Guo et al 2013	Liang et al 2013	Sukno et al 2015 (clinical dataset group)	De Jong et al 2018	Vezzetti et al 2018	Abu et al 2019	Wang et al 2019	Baksi et al 2020	Bannister et al 2020	Min	Max
	No of LM	17	20	14	21	13	10	23	21	12		
14	**Alar curvature point (left)**			2.01	1.70			2.46			1.70	2.46
15	**Subalare (right)**		2.36		1.30						1.30	2.36
16	**Subalare (left)**		1.59		1.50						1.50	1.59
17	**Crista philtri (right)**		2.31						2.77		2.31	2.77
18	**Crista philtri (left)**		1.99						3.81		1.99	3.81
19	**Chelion (right)**	0.99	3.08	2.55	1.80		−0.56		1.93	2.20	0.99	3.08
20	**Chelion (left)**	1.01	3.08	2.55	1.40				3.35	2.00	1.01	3.35
21	**Pogonion**	1.81		3.35	2.00			1.58	3.50		1.58	3.50
22	**Gnathion**		5.31					1.89	4.85	2.90	1.89	5.31
23	**Glabella**				3.00			2.87	6.35		2.87	6.35

Table 5: Landmark identification error for 23 of the most reported landmarks in the included studies.

155x105mm (300 x 300 DPI)

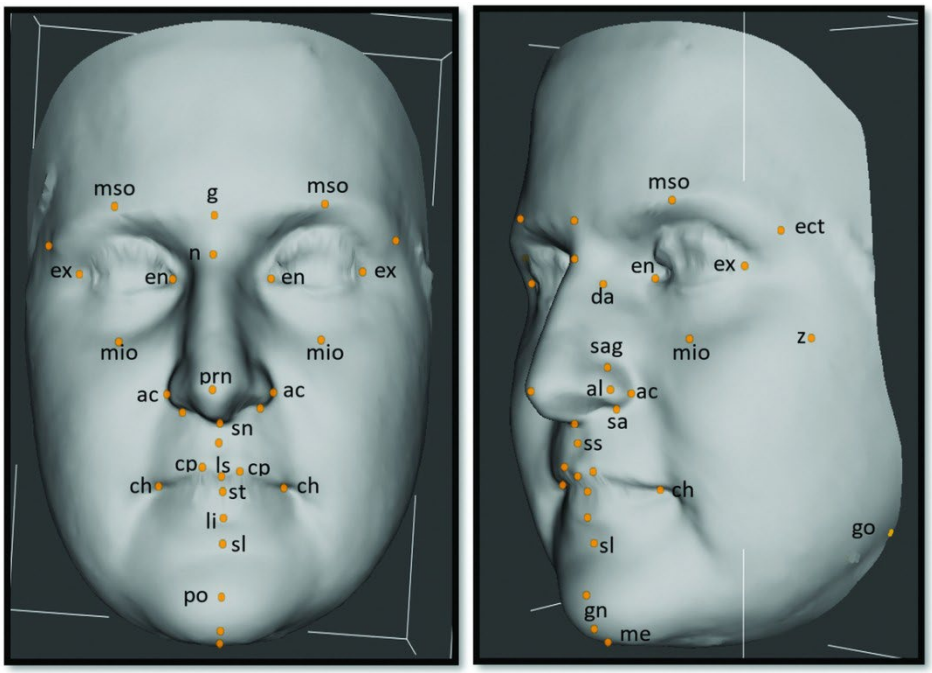


Figure 2: The 3D facial landmarks that have been used in the included studies. Midline landmarks: g=glabella, n=nasion, prn=pronasale, sn=subnasale, ls=labialesuperius, st=stomion, li=labialeinferius, sl=sublabiale, po=pogonion, g=gnathion, me=menton. Bilateral landmarks: en=endocanthion, ex=xocanthion, mso= mid-supraorbital, mio=mid-infraorbital, da=dacryon, al=alare, ac= alar curvature, sag=superior alar groove, sa=subalare, ect=ectoconchion, ss=subspinale, cp= crista philtri, ch=chelion, sl=sublabiale z= zygion. (Tragion and earlobe tip landmarks are not illustrated as the ears area is not shown in this 3D model).

104x74mm (300 x 300 DPI)

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Main file: P 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Main file: P1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Main file: P2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Main file: P2,3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Main file: P3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Main file: P4
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Sup file: table 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Main file: P4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Main file: P4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Main file: P4,5 Table3,4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Main file: P4,5 Table3,4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Main file: P5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Main file: P5

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Main file: P5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Main file: P5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Main file: P5,6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	
Study characteristics	17	Cite each included study and present its characteristics.	Main file: P 7-11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Main file: P5
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 4,5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Table 3,4
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Main file: P11-14
	23b	Discuss any limitations of the evidence included in the review.	Main file:

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
			P14-15
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	Main file: P15
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Main file: P3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Main file: P3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Main file: P16
Competing interests	26	Declare any competing interests of review authors.	Main file: P16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

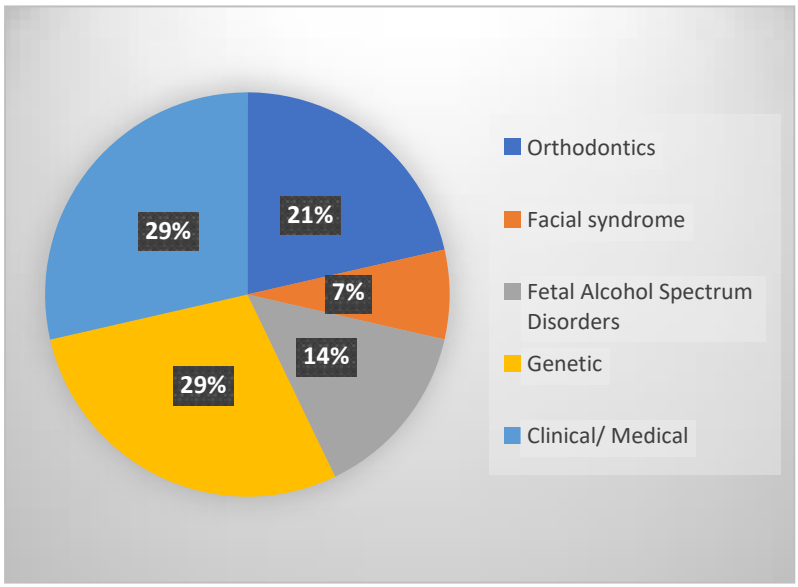
For more information, visit: http://www.prisma-statement.org/

Supplementary table 1: Search strategy used in this study.

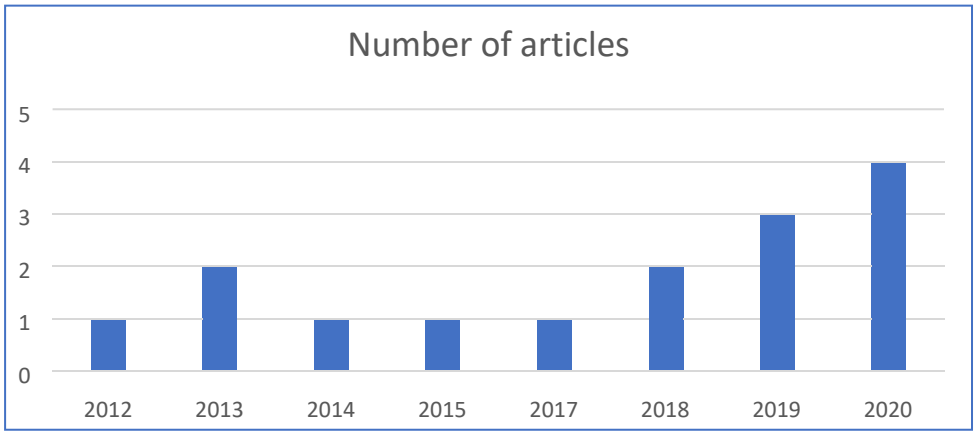
1.	Landmarking
2.	Landmarks
3.	Anatomic landmarks (MeSH)
4.	**1 OR 2 OR 3**
5.	Face / diagnostic imaging (MeSH)
6.	Face
7.	Facial
8.	**5 OR 6 OR 7**
9.	3D
10.	Three dimensional
11.	Three-dimensional
12.	Imaging/Three-dimensional (MeSH)
13.	**9 OR 10 OR 11 OR 12**
14.	Valid*
15.	Accuracy
16.	Accurate
17.	Repeatability
18.	Agreement
19.	Concordance
20.	Reproducibility
21.	Reliability
22.	Compar*
23.	Performance
24.	Error
25.	Reproducibility of results (MeSH)
26.	Observer variation (MeSH)
27.	**14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23 OR 24 OR 25 OR 26**
28.	Automat*
29.	Detection
30.	Identification
31.	Localisation
32.	Algorithm
33.	**28 OR 29 or 30 or 31 or 32**

34.	Humans (MeSH)
35.	Human*
36.	Male*
37.	Female*
38.	**34 OR 35 OR 36 OR 37**
39.	**4 AND 8 AND 13 AND 27 AND 33 AND 38**

Supplementary Figure 1: Research topic area of the included studies.



Supplementary Figure 2: Number of included studies by year of publication.



Supplementary table 2: 2mm threshold of landmark identification error.

2mm thershold		Guo et al 2013	Liang et al 2013	Sukno et al 2015 (clinical dataset group)	De Jong et al 2018	Vezzetti et al 2018	Abu et al 2019	Wang et al 2019	Baksi et al 2020	Bannister et al 2020
	No of LM	**17**	**20**	**14**	**21**	**13**	**10**	**23**	**21**	**12**
1	**Nasion**	≤2mm	>2mm	≤2mm	≤2mm	>2mm	≤2mm	>2mm	>2mm	>2mm
2	**Pronasale**	≤2mm	≤2mm	≤2mm	≤2mm	>2mm	≤2mm		≤2mm	≤2mm
3	**Subnasale**	≤2mm	>2mm	≤2mm	≤2mm	>2mm	>2mm	>2mm	≤2mm	>2mm
4	**Labiale superius**	≤2mm	>2mm	>2mm	≤2mm		>2mm	>2mm	≤2mm	>2mm
5	**Stomion**	≤2mm	≤2mm				>2mm		≤2mm	
6	**Labiale inferius**	≤2mm	>2mm	>2mm	>2mm		>2mm	≤2mm	>2mm	>2mm
7	**Endocanthion (right)**	≤2mm	>2mm	≤2mm	≤2mm	>2mm			>2mm	>2mm
8	**Endocanthion (left)**	≤2mm	>2mm	≤2mm	≤2mm	>2mm			>2mm	>2mm
9	**Exocanthion (right)**	≤2mm	>2mm	>2mm	≤2mm	>2mm			>2mm	>2mm
10	**Exocanthion (left)**	≤2mm	>2mm	>2mm	≤2mm	>2mm			>2mm	≤2mm
11	**Alare(R)**	≤2mm	≤2mm			>2mm	≤2mm		>2mm	
12	**Alare(L)**	≤2mm	>2mm			>2mm			>2mm	
13	**Alar curvature point (right)**			>2mm	≤2mm			>2mm		
14	**Alar curvature point (left)**			>2mm	≤2mm			>2mm		
15	**Subalare (right)**		>2mm		≤2mm					
16	**Subalare (left)**		≤2mm		≤2mm					
17	**Crista philtri (right)**		>2mm						>2mm	
18	**Crista philtri (left)**		≤2mm						>2mm	
19	**Chelion (right)**	≤2mm	>2mm	>2mm	≤2mm		≤2mm		≤2mm	>2mm
20	**Chelion (left)**	≤2mm	>2mm	>2mm	≤2mm				>2mm	>2mm
21	**Pogonion**	≤2mm		>2mm	≤2mm			≤2mm	>2mm	
22	**Gnathion**		>2mm					≤2mm	>2mm	>2mm
23	**Glabella**				>2mm			>2mm	>2mm	

Supplementary table 3: 1mm threshold of landmark identification error.

1mm threshold		Guo et al 2013	Liang et al 2013	Sukno et al 2015 (clinical dataset group)	De Jong et al 2018	Vezzetti et al 2018	Abu et al 2019	Wang et al 2019	Baksi et al 2020	Bannister et al 2020
	No of LM	17	20	14	21	13	10	23	21	12
1	Nasion	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm
2	Pronasale	≤1mm	>1mm	>1mm	>1mm	>1mm	>1mm		>1mm	>1mm
3	Subnasale	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm	>1mm
4	Labiale superius	>1mm	>1mm	>1mm	>1mm		>1mm	>1mm	>1mm	>1mm
5	Stomion	>1mm	>1mm				>1mm		>1mm	
6	Labiale inferius	>1mm	>1mm	>1mm	>1mm		>1mm	>1mm	>1mm	>1mm
7	Endocanthion (right)	>1mm	>1mm	>1mm	>1mm	>1mm			>1mm	>1mm
8	Endocanthion (left)	>1mm	>1mm	>1mm	>1mm	>1mm			>1mm	>1mm
9	Exocanthion (right)	>1mm	>1mm	>1mm	>1mm	>1mm			>1mm	>1mm
10	Exocanthion (left)	>1mm	>1mm	>1mm	>1mm	>1mm			>1mm	>1mm
11	Alare(R)	>1mm	>1mm			>1mm	≤1mm		>1mm	
12	Alare(L)	>1mm	>1mm			>1mm			>1mm	
13	Alar curvature point (right)			>1mm	>1mm			>1mm		
14	Alar curvature point (left)			>1mm	>1mm			>1mm		
15	Subalare (right)		>1mm		>1mm					
16	Subalare (left)		>1mm		>1mm					
17	Crista philtri (right)		>1mm						>1mm	
18	Crista philtri (left)		>1mm						>1mm	
19	Chelion (right)	≤1mm	>1mm	>1mm	>1mm		≤1mm		>1mm	>1mm
20	Chelion (left)	>1mm	>1mm	>1mm	>1mm				>1mm	>1mm
21	Pogonion	>1mm		>1mm	>1mm			>1mm	>1mm	
22	Gnathion		>1mm					>1mm	>1mm	>1mm
23	Glabella				>1mm			>1mm	>1mm	

Supplementary table 4: 0.5mm threshold of landmark identification error.

0.5mm threshold		Guo et al 2013	Liang et al 2013	Sukno et al 2015 (clinical dataset group)	De Jong et al 2018	Vezzetti et al 2018	Abu et al 2019	Wang et al 2019	Baksi et al 2020	Bannister et al 2020
	No of LM	**17**	**20**	**14**	**21**	**13**	**10**	**23**	**21**	**12**
1	**Nasion**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm
2	**Pronasale**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm		>0.5mm	>0.5mm
3	**Subnasale**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm
4	**Labiale superius**	>0.5mm	>0.5mm	>0.5mm	>0.5mm		>0.5mm	>0.5mm	>0.5mm	>0.5mm
5	**Stomion**	>0.5mm	>0.5mm				>0.5mm		>0.5mm	>0.5mm
6	**Labiale inferius**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm
7	**Endocanthion (right)**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm			>0.5mm	>0.5mm
8	**Endocanthion (left)**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm			>0.5mm	>0.5mm
9	**Exocanthion (right)**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm			>0.5mm	>0.5mm
10	**Exocanthion (left)**	>0.5mm	>0.5mm	>0.5mm	>0.5mm	>0.5mm			>0.5mm	>0.5mm
11	**Alare(R)**	>0.5mm	>0.5mm			>0.5mm	>0.5mm		>0.5mm	
12	**Alare(L)**	>0.5mm	>0.5mm			>0.5mm	>0.5mm		>0.5mm	
13	**Alar curvature point (right)**			>0.5mm	>0.5mm			>0.5mm		
14	**Alar curvature point (left)**			>0.5mm	>0.5mm			>0.5mm		
15	**Subalare (right)**		>0.5mm		>0.5mm					
16	**Subalare (left)**		>0.5mm		>0.5mm					
17	**Crista philtri (right)**		>0.5mm						>0.5mm	
18	**Crista philtri (left)**		>0.5mm						>0.5mm	
19	**Chelion (right)**	>0.5mm	>0.5mm	>0.5mm	>0.5mm		>0.5mm		>0.5mm	>0.5mm
20	**Chelion (left)**	>0.5mm	>0.5mm	>0.5mm	>0.5mm				>0.5mm	>0.5mm
21	**Pogonion**	>0.5mm		>0.5mm	>0.5mm			>0.5mm	>0.5mm	
22	**Gnathion**		>0.5mm					>0.5mm	>0.5mm	>0.5mm
23	**Glabella**				>0.5mm			>0.5mm	>0.5mm	

Supplementary table 5: Performance metrics used for the evaluation of the reference standard and the index test (Research Q:.accuracy of automated 3D facial landmarking when compared to manual landmarking).

Author/ year	Metrics used for evaluation of reference standard (ground truth)	evaluation of automated landmarking method		
		performance metrics used for evaluation	individually assessment in each of the three dimensions—x-axis, y-axis, and z-axis	overall v induvial landmark evaluation
Baksi et al 2020	Intraclass correlation (ICC) was used to assess intra and inter-observer reliability.	Difference in landmarks identified manually compared with those identified by the automated algorithm. (Mean± SD)	Yes	Overall+ individual landmark evaluation
Bannister et al. 2020	Inter and intra-observer error evaluation. The mean and standard deviation of the Euclidean distances between different sets of landmarks.	The mean and standard deviation of the Euclidean distances between different sets of landmarks. (Mean± SD)	No	Overall+ individual landmark evaluation
Fu et al 2020	No	1- Mean error (overall only) 2- Area-Under-the-Curve (AUC) 3- Failure Rate (FR) > 1.2	No	only overall evaluation.
Wang et al. 2019	No	Direct comparison of manual and automatic landmark placements. (Mean only)	Yes (plots only)	Overall+ individual landmark evaluation
Abu et al 2019	The ICC test was performed to check the reliability of the linear measurements data taken manually.	Difference in landmarks identified manually compared with those identified by the automated algorithm. (Mean± SD)	No	Overall+ individual landmark evaluation
Huang et al 2019	No	1- The localization accuracy was evaluated by calculating the Euclidean distance between the ground truth (GT) and the predictions in 2D. 2- Normalization mean error 3D (Mean± SD)	No	only overall evaluation.
De Jong et al 2018	No	Cross-validation (leave-one-out experiments). The localization accuracy was evaluated by calculating the Euclidean distance between the ground truth (training data) in mm and the predicted automated points.	No	Overall+ individual landmark evaluation
Li et al. 2017	The analysis of measurement error and observer error for manual landmarking was done by comparing the overall shape variation among groups by using a multivariate analysis of variance (MANOVA).	Analyses of morphological variation were performed using geometric morphometric methods, and direct comparisons of methods followed the Bland–Altman approach. Comparison of mean shape variation among groups by using multivariate analysis of variance (MANOVA) and permutation of the Procrustes distances.	No	No clear figure was reported. Mean error of individual LM was only presented in plots.
Sukno et al. 2015	No	Direct comparison of manual and automatic landmark placements. (Mean± SD)	No	Overall+ individual landmark evaluation
Guo et al 2013	No	Direct comparison of manual and automatic landmark placements. The mean and standard deviation (SD) of the annotation errors measured in Euclidean distance, as well as the root mean square (RMS) errors.	No	Overall+ individual landmark evaluation
Liang et al 2013	No	Direct comparison of manual and automatic landmark placements. (Mean± SD)	No	Overall+ individual landmark evaluation
Vezzetti et al. 2012	No	Direct comparison of manual and automatic landmark placements. (Mean only)	No	Overall+ individual landmark evaluation

Supplementary table 6: Performance metrics used for the evaluation of the reliability of clinical or genetic facial analysis. (research Q2).

Author/ year	The reliability of further analysis (genetic or clinical) obtained through automatic landmarking when compared to those based on manual landmarking.	
	Type of analysis	Statistical test used for the analysis
Baksi et al 2020	Evaluation of the reliability of ten anthropometric linear measurements for clinical use.	ICC was also applied to compare agreement between manual and automated landmarks for the 10 anthropometric linear measurements. ICC values less than 0.5 were deemed indicative of poor reliability, values between 0.5 and 0.75 of moderate reliability, values between 0.75 and 0.9 indicative of good reliability, and values greater than 0.90 to be excellent reliability.
Palmer et al 2020	Direct comparison of eight linear measurements obtained by manual (ground truth), indirect (non-expert assessor) and automated (cliniface) methods.	Paired samples t-testing with $\alpha = 0.05$ was used to evaluate that the automatically generated measurements of Cliniface are not significantly different in accuracy to those taken by the expert.
Abu et al 2019	evaluation of the validity of the automated methods against the manual landmarking by comparing **the average distance of all the eight linear** measurements extracted by both methods	2-tailed paired t-test with the mean difference, $\mu > 0$ and significance value, $p = 0.05$.
Huang et al 2019	Evaluation of the accuracy of the automated generated palpebral fissure length (PFL) measurements derived from 3D facial images which can be used in assisting the diagnosis of fetal alcohol spectrum disorder	Palpebral fissure length (PFL) mean error.
De Jong et al 2018	Study used twin-based heritability information to perform heritability analysis for landmarking accuracy.	A narrow sense heritability estimate for a number of features derived from landmark coordinates.
Li et al. 2017	The automated method was validated by performing variation patterns in large dataset of automated and manual landmark data to determine if the variance structure produced by the two methods was similar.	variation structure between the manual and the automated landmark was assessed by using partial least squares (PLS) regression to compare covariation structure between the two methods.
Lippold et al 2014	Validate the performance of the automated method by determining the difference of each of seven linear facial measurements.	Lin's concordance correlation coefficient CCC was used for the evaluation of equivalence between methods and concordance of continuous data.

Supplementary Table7: List of excluded studies at full-text review with reasons for exclusion.

Reference	Reason for exclusion
Ferrario et al. 1994 (1)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Ferrario et al. 1995(2)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Mishima et al. 1996 (3)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Ferrario et al. 1998(4)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Yamada et al. 1999(5)	No comparison to reference
Yamada et al. 1999(6)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Xu et al. 1999(7)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Ferrario et al. 1999(8)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Hajeer et al. 2002(9)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Yamada et al. 2002(10)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Naftel et al. 2004(11)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Akakin and Sankur 2007(12)	Not a medical or biological study
Zhang et al.(13)	Article is not in English
Sattae and Seguier 2009(14)	Not a medical or biological study
Galantucci et al. 2009(15)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Deli et al. 2010(16)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Wu and Chen 2010(17)	Not a medical or biological study
Claes et al. 2011(18)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Calignano and Vezzetti 2011(19)	more recent algorithm by the same research team
Zaho et al. 2011(20)	Not a medical or biological study
Liberton et al. 2011	No access to full article
Wei et al 2011(21)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Kung et al. 2012 (22)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)

Loconsole et al. 2012(23)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Vezzetti et al. 2012(24)	more recent algorithm by the same research team
Peng et al. 2013(25)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Mercan et al. 2013(26)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Berssenbrügge et al 2014(27)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Xiong et al. 2015 (28)	Article is not in English
Tse et al. 2015	No access to full article
Feng et al. 2015(29)	Not a medical or biological study
Chendab et al. 2015(30)	No quantitative analysis for the clinical set.
Zhao et al. 2016(31)	Not a medical or biological study
Liu et al. 2015(32)	Not a medical or biological study
Guo et al. 2016(33)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Jurda and Urbanova 2016(34)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Danelakis et al. 2016(35)	Not a medical or biological study
Caple et al. 2016(36)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
De Jong et al. 2015(37)	more recent algorithm by the same research team
Liu and Cohen 2017(38)	Not a medical or biological study
Liu et al 2017(39)	Not a medical or biological study
Liu et al. 2017(40)	Not a medical or biological study
Tsagkrasoulis et al. 2017(41)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Bannister et al 2017	No access to full article
Gilani et al. 2017(42)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Hoskens et al. 2018(43)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Ekrami et al. 2018(44)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Tuin et al. 2019(45)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Yasuda et al. 2019(46)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Gul et al 2019(47)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)

Ahmedt-Aristizabal et al. 2019(48)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Macedo et al 2019(49)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Quintana et al. 2019(50)	Not a medical or biological study
Akbarian et al. 2019(51)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Abbas et al. 2019(52)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Suttie et al. 2019.	No access to full article
Tuin et al. 2020(53)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Agbolade et al. 2020(54)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Zhu et al 2020(55)	Article is not in English
Guo et al 2020(56)	Article is not in English
Lum et al. 2020(57)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Lv et al. 2020(58)	Not a medical or biological study
Hong et al. 2020(59)	Not a medical or biological study
Dai et al. 2020(60)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Nazri et al. 2020(61)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)
Zhu et al 2020(62)	Study is not relevant to the review topic (Evaluation of automated 3D facial landmarking in human)

List of excluded articles related to table 7:

1. Ferrario V, Sforza C, Serrao G, Puletto S, Bignotto M, Tartaglia G. Comparison of soft tissue facial morphometry in children with Class I and Class II occlusions. The International Journal of Adult Orthodontics and Orthognathic Surgery. 1994;9(3):187-94.
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